Structural Self-Consistency for Quality Estimation in LLM-Based Information Extraction

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Abstract

Large language models increasingly tackle structured information extraction (IE) via multi-sample generation, yet practitioners lack principled guidance on which confidence signals to trust—and for what purpose. We systematically compare five unsupervised confidence signals spanning four families—model confidence (log-probability), surface agreement (Fleiss' κ, exact match), set-aware structural similarity (Soft Jaccard), and entity-level majority voting—for named entity recognition (NER) and relation extraction (RE) across two models (Qwen3-8B, LLaMA-3.1-8B) and two datasets (SciERC, CoNLL-2003). We report three findings. **(1)** Structural consistency signals provide the strongest unsupervised quality estimation: entity voting achieves AUROC 0.73 and ρ=0.50 on NER, substantially outperforming log-probability (AUROC 0.56, ρ=0.14). **(2)** A *correlation–selection gap* exists: despite strong instance-level correlation, no unsupervised signal beats greedy decoding for NER best-of-N selection; only log-probability enables selection on RE (+2.4pp). **(3)** All signals degrade on relational tasks due to compound error propagation, but remain valid within the model's capability range. These results hold across a 2×2 model–dataset matrix ($|\Delta\rho| \leq 0.006$ cross-model) and three random seeds ($\sigma_\rho \leq 0.008$). Our findings disentangle quality estimation from sample selection and provide actionable guidance for deploying multi-sample IE systems.

1 Introduction

Multi-sample generation—drawing N candidate outputs from a language model and selecting or aggregating among them—has become a standard tool for improving LLM performance on reasoning tasks [Wang et al.(2023), Snell et al.(2024), Brown et al.(2024)]. For structured information extraction (IE), where outputs are entity sets or relation tuples, the natural question is: *which confidence signal should guide the selection or abstention decision?*

This question is more nuanced than it appears. Prior work on self-consistency [Wang et al.(2023)] uses exact-match voting, which conflates two distinct capabilities: (i) *quality estimation*—predicting which inputs yield reliable outputs—and (ii) *sample selection*—choosing the best output among N candidates. For structured outputs, surface-level agreement metrics further suffer from semantic blindness: two entity sets may differ in surface form yet agree structurally [Nédellec and Xia(2026)].

We address these gaps with a systematic comparison of five confidence signals across four families for entity-level IE (Section 3). Our study spans two tasks (NER, RE), two models (Qwen3-8B, LLaMA-3.1-8B), two datasets (SciERC, CoNLL-2003), and evaluates both quality estimation (Spearman ρ, Kendall τ, AUROC) and best-of-N selection.

Our contributions are:

1. We show that structural consistency signals—entity voting confidence and Soft Jaccard—are the strongest unsupervised quality estimators for IE, outperforming log-probability by 17–30 AUROC points and 0.2–0.4 in Spearman ρ (Section 5.1).

2. We identify a *correlation–selection gap*: strong instance-level quality prediction does not translate to effective per-sample selection. On NER, no signal surpasses greedy decoding despite 16pp of oracle headroom (Section 5.3).
3. We characterize a *task-complexity boundary*: compound error propagation in relational tasks degrades all consistency signals on the full instance set, but conditional filtering on the model's capability range restores discriminative power (Section 5.4).
4. We validate these findings across a 2×2 model–dataset matrix with cross-model $|\Delta\rho| \leq 0.006$ and three-seed $\sigma_\rho \leq 0.008$ (Section 5.2).

2 Related Work

Self-consistency for reasoning. Wang et al. [Wang et al.(2023)] introduced self-consistency as majority voting over chain-of-thought samples, demonstrating gains on arithmetic and commonsense reasoning. Subsequent work has extended this to confidence-weighted voting (CISC; cisc2025), reliability-aware adaptive thresholds (ReASC; reasc2026), sub-claim decomposition (Atomic Self-Consistency; atomic2024), ranked voting [Ranked Voting(2025)], and KL-divergence-based reward-free selection [Self-Certainty(2026)]. All of these target discrete-answer reasoning tasks; none addresses the set-structured outputs characteristic of IE.

Uncertainty quantification for LLMs. Kuhn et al. [Kuhn et al.(2023)] proposed semantic entropy over meaning-equivalent clusters. Surveys by uqsurvey2025 and uqcalibsurvey2025 provide comprehensive taxonomies. Semantic entropy probes [Semantic Probes(2024)] and latent self-consistency [Latent SC(2025)] reduce computational cost via embedding-space approximations. Manakul et al. [Manakul et al.(2023)] use sampling-based self-checking for hallucination detection. CoCoA [CoCoA(2025)] combines confidence and consistency via minimum Bayes risk; Vashurin et al. [Vashurin et al.(2025)] systematically evaluate four UQ methods across tasks—none including structured IE. Our work fills this gap by comparing confidence signal families specifically for entity-level and relational IE.

Calibration and selective prediction. Model calibration has been studied extensively [Guo et al.(2017), Kadavath et al.(2022), Desai and Durrett(2020)]. Selective prediction—abstaining on uncertain inputs—is surveyed by geifman2017selective and knowyourlimits2025. The confidence estimation survey of confestsurvey2024 provides a broad taxonomy. For structured prediction, Kuleshov & Liang [Kuleshov and Liang(2015)] and Jagannatha & Yu [Jagannatha and Yu(2022)] address calibration but not multi-sample consistency. Our quality estimation results (Section 5.1) directly apply to selective prediction / abstention.

Confidence in information extraction. Culotta & McCallum [Culotta and McCallum(2004)] studied confidence estimation for CRF-based IE. For NER, token-level uncertainty methods include MC-dropout [Gal and Ghahramani(2016), Xiao and Wang(2019)] and deep ensembles [Lakshminarayanan et al.(20 Shelmanov et al.(2021)]. Nédellec & Xia [Nédellec and Xia(2026)] recently studied consistency–accuracy correlation for hard-prompted LLMs on plant-health IE, but used surface-level string matching (which they acknowledge "overlooks semantically consistent outputs"), a single domain, and did not study best-of-N selection. Our Soft Jaccard metric directly addresses their limitation, and our multi-signal comparison provides a more complete picture.

Test-time scaling and sample selection. Snell et al. [Snell et al.(2024)] and Brown et al. [Brown et al.(2024)] formalize test-time compute scaling. Verifier and reward models [Cobbe et al.(2021), Lightman et al.(2023)] provide supervised selection signals. Our work studies the unsupervised regime where such verifiers are unavailable, and identifies a fundamental gap between quality estimation and selection (Section 5.3). Conformal prediction approaches [Fisch et al.(2021), Angelopoulos et al.(2021)] provide coverage guarantees but require held-out calibration data; our signals are fully unsupervised.

3 Method

3.1 Problem Formulation

Given an input x, a language model $\mathcal{M}$ generates N candidate structured outputs $\{y_1, \ldots, y_N\}$ via temperature sampling ($T{>}0$). Each output y_i is a set of typed entities (NER) or relation tuples (RE). A greedy reference y_0 is obtained via $T{=}0$. We study two problems: (i) *quality estimation*—predicting instance-level output quality $q(x) = F_1(y_0, y^*)$ where y^* is the gold standard, and (ii) *sample selection*—choosing $\hat{y} = \arg\max_{y_i} s(y_i)$ to maximize $F_1(\hat{y}, y^*)$.

3.2 Confidence Signals

We compare five signals spanning four families:

Log-probability (logprob). The mean token-level log-probability of the generated sequence: $\textsc{logprob}(y_i) = \frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \log p_\theta(y_i^{(t)} \mid x, y_i^{(<t)})$. This is the only *per-sample* signal: it scores each y_i independently. For quality estimation, we use $\textsc{logprob}(x) = \textsc{logprob}(y_0)$, the greedy output's log-probability.

Exact Match (em). The fraction of sample pairs producing identical outputs: $\textsc{em}(x) = \binom{N}{2}^{-1} \sum_{i<j} \mathbb{1}[y_i = y_j]$.

Fleiss' Kappa (fk). Token-level inter-annotator agreement across N samples, treating each sample as an annotator assigning BIO tags.

Soft Jaccard (sj). A set-aware structural similarity metric. For a pair of entity sets (E_a, E_b): (i) group entities by type; (ii) within each type t, compute the optimal bipartite assignment via Hungarian matching using span-overlap similarity; (iii) normalize by $\max(|E_a^t|, |E_b^t|)$ to obtain $\textsc{sj}_t$; (iv) average across types. The instance-level score is $\textsc{sj}(x) = \binom{N}{2}^{-1} \sum_{i<j} \textsc{sj}(y_i, y_j)$. SJ generalizes to arbitrary set-structured outputs (relations, events) without task-specific design.

Entity Voting Confidence (vote). For each entity e in the greedy output y_0, compute $v(e) = |\{i : e \in y_i\}|/N$ (the fraction of samples containing e). The instance-level score is $\textsc{vote}(x) = |y_0|^{-1} \sum_{e \in y_0} v(e)$. Unlike SJ, voting is entity-specific and does not generalize to arbitrary structures.

3.3 Evaluation Metrics

We evaluate quality estimation via Spearman ρ, Kendall τ, and AUROC (binarized at median F_1) between signal scores and instance-level F_1. We evaluate sample selection via best-of-N F_1:

Table 1: Quality estimation on SciERC (Qwen3-8B, N=8, T=1.0). Spearman ρ, Kendall τ, and AUROC between each signal and instance-level F_1. Best in **bold**, second underlined.

	NER (n=529)			**RE** (n=397)	
Signal	ρ	τ	AUROC	ρ	AUROC
VOTE	**0.502**	**0.377**	**0.728**	**0.422**	**0.716**
SJ	0.357	0.266	0.666	0.211	0.582
FK	0.265	0.203	0.624	0.253	0.631
EM	0.257	0.192	0.623	0.100	0.504
LOGPROB	0.140	0.101	0.559	0.247	0.635

$F_1(\arg\max_{y_i} s(y_i), y^*)$, macro-averaged across instances. We report results on the *full* set (all instances) and the *conditional* set (instances where $F_1(y_0, y^*) > 0$), which separates model capability errors from uncertainty estimation.

4 Experimental Setup

Datasets. **SciERC** [Luan et al.(2018)]: scientific abstracts with 6 entity types and 7 relation types; we use the standard test set (n=529 for NER, n=397 for RE). **CoNLL-2003** [Tjong Kim Sang and De Meulc news text with 4 entity types (PER, ORG, LOC, MISC); test set n=3,453 (n=2,756 valid after filtering). **WNUT-2017** [Derczynski et al.(2017)]: social media text with 6 entity types; used only for out-of-domain evaluation (n=1,287).

Models. **Qwen3-8B** [Qwen Team(2025)]: fine-tuned with LoRA on SciERC and CoNLL-2003 training sets. **LLaMA-3.1-8B-Instruct** [LLaMA Team(2024)]: fine-tuned with LoRA on SciERC and CoNLL-2003. Both models are 8B-parameter instruction-tuned models representing distinct architecture families.

Generation. N=8 samples at T=1.0 with greedy (T=0) reference, seed 42. For robustness analysis, we also evaluate N=16 with three seeds (42, 123, 456) on LLaMA × CoNLL.

5 Results

5.1 Quality Estimation

Table 1 presents quality estimation results on SciERC. On NER, entity voting confidence achieves the strongest correlation (ρ=0.502, AUROC=0.728), followed by Soft Jaccard (ρ=0.357, AUROC=0.666). Both structural consistency signals substantially outperform log-probability (ρ=0.140, AUROC=0.559), with voting yielding a +0.362 ρ advantage and +16.9 AUROC points.

On RE, voting again leads (ρ=0.422, AUROC=0.716), but the margins narrow: FK (ρ=0.253) and logprob (ρ=0.247, AUROC=0.635) become competitive on specific metrics. EM is weakest on RE (ρ=0.100), confirming that exact-match agreement is too strict for structured outputs with compositional variation.

Conditional analysis. Filtering to instances where $F_1(y_0) > 0$ (NER n=501; RE n=296) improves all signals. On NER, SJ rises from ρ=0.357 to 0.412 while voting moves from 0.502 to

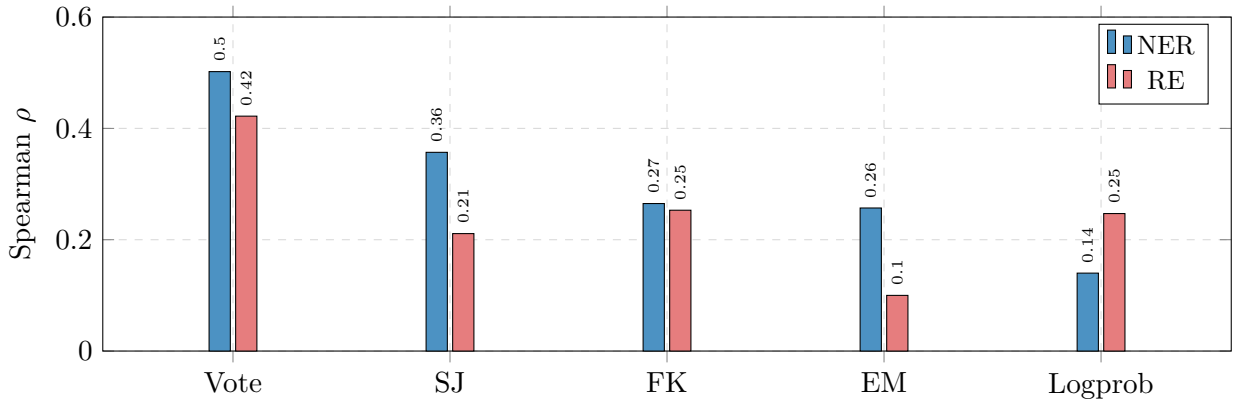


Figure 1: Spearman ρ between confidence signals and instance-level F_1 on SciERC (Qwen3-8B, N=8). Structural signals (Vote, SJ) dominate on NER; on RE, logprob becomes competitive due to compound error degradation of consistency signals.

Table 2: 2×2 cross-model × cross-dataset quality estimation (Spearman ρ, full set, N=8). Structural signals outperform logprob in all four configurations. Cross-model $|\Delta\rho| \leq 0.006$ on the same dataset.

Configuration	SJ	VOTE	FK	EM	LOGPROB	n
Qwen × SciERC	0.357	**0.502**	0.265	0.257	0.140	529
Qwen × CoNLL	0.436	0.424	0.400	**0.451**	0.316	2756
LLaMA × SciERC	0.363	—	0.327	—	0.183	529
LLaMA × CoNLL	<u>0.431</u>	0.426	0.428	0.431	0.307	2756

0.450—SJ's relative position improves in the conditional regime. On RE, the conditional lift is dramatic: SJ rises from 0.211 to 0.368 (+0.157), voting from 0.422 to 0.463, while logprob improves only modestly (0.247 → 0.146—in fact decreasing, as logprob conflates capability gaps with uncertainty). This motivates the task-complexity analysis in Section 5.4.

5.2 Cross-Model and Cross-Dataset Generalization

To assess generalization, we evaluate all signals across a 2×2 matrix of models (Qwen3-8B, LLaMA-3.1-8B) and datasets (SciERC, CoNLL-2003).

Spearman ρ (Table 2). Structural signals outperform logprob in all four configurations (structural > logprob rate = 100%). Cross-model consistency is remarkably high: on CoNLL, Qwen SJ ρ=0.436 vs. LLaMA SJ ρ=0.431 (Δ=0.005); on SciERC, Qwen SJ ρ=0.357 vs. LLaMA ρ=0.363 (Δ=−0.006). Both models exhibit the same dataset effect (CoNLL > SciERC), confirming that consistency signal strength is a task property, not a model artifact.

AUROC (Table 3). Structural signals achieve the highest AUROC in 3 of 4 configurations. The exception is LLaMA × CoNLL, where logprob AUROC (0.761) exceeds structural signals (SJ 0.704, Vote 0.709). This occurs in the high-baseline regime (greedy F_1=0.927): when model performance is near-perfect, log-probability calibration is tight and threshold-based classification (AUROC) favors the well-calibrated signal. However, logprob's *ranking* ability (ρ=0.307) remains substantially below

Table 3: AUROC across the 2×2 matrix (N=8). Structural signals lead in 3 of 4 configurations; logprob is competitive only on CoNLL (high-baseline regime with well-calibrated models).

Configuration	SJ	VOTE	LOGPROB	Best signal
Qwen × SciERC	0.666	**0.728**	0.559	Vote 0.728
Qwen × CoNLL	**0.801**	0.789	0.753	SJ 0.801
LLaMA × SciERC	0.702	**0.718**	0.558	Vote 0.718
LLaMA × CoNLL	0.704	0.709	**0.761**	Logprob 0.761



Figure 2: Cross-model consistency of Soft Jaccard ρ across datasets. The two architecturally distinct models produce near-identical quality estimation performance ($|\Delta\rho| \leq 0.006$), demonstrating that structural consistency is a model-agnostic property.

structural signals (ρ=0.431). This AUROC–ρ divergence illustrates why multiple evaluation metrics are necessary: AUROC measures threshold classification while ρ measures rank discrimination.

Seed robustness. On LLaMA × CoNLL with N=16 and three seeds (42, 123, 456), SJ ρ is 0.463 ± 0.008, voting ρ is 0.458 ± 0.003, and logprob ρ is 0.310 ± 0.005 (Table 4). The structural–logprob gap is stable across all seeds (structural beats logprob rate = 100%). The N=8 → N=16 improvement is modest for SJ (+0.033), suggesting near-saturation on CoNLL NER.

5.3 The Correlation–Selection Gap

Table 5 and Fig. 3 reveal the central finding of this work. On NER, despite voting's strong quality estimation (ρ=0.502, AUROC=0.728), its best-of-N selection (F_1=0.596) *underperforms* greedy decoding (F_1=0.609). All consistency signals fail to beat greedy, while the oracle achieves 0.769 (+16pp headroom).

This gap arises from a fundamental distinction. Consistency signals are *instance-level aggregates*: they assign one score per input x by measuring agreement across all N samples. They predict *which inputs are easy vs. hard*—ideal for selective prediction and abstention. But they cannot rank individual samples y_i against each other, which is required for best-of-N selection.

Log-probability is the only *per-sample* signal: it scores each y_i independently. On RE, where greedy is suboptimal (F_1=0.374, oracle 0.561), logprob enables effective selection (+2.4pp over greedy). On NER, where greedy is near-optimal, even logprob provides no benefit.

Practical implications. For selective prediction (deciding whether to trust an output), structural consistency signals are the tool of choice. For best-of-N selection (choosing the best output),

Table 4: Three-seed robustness (LLaMA × CoNLL, N=16). All structural signals consistently outperform logprob with low variance.

Metric	Seed 42	Seed 123	Seed 456	Mean ± Std
SJ ρ	0.464	0.470	0.455	0.463 ± 0.008
FK ρ	0.457	0.459	0.448	0.455 ± 0.006
Logprob ρ	0.306	0.316	0.308	0.310 ± 0.005
Vote ρ	0.455	0.461	0.457	0.458 ± 0.003
SJ AUROC	0.739	0.740	0.729	0.736 ± 0.006

Table 5: Best-of-N selection on SciERC (N=8, T=1.0). Despite strong quality estimation, no signal beats greedy on NER. On RE, only logprob surpasses greedy (+2.4pp), as the sole per-sample signal.

Signal	NER F_1	RE F_1
Oracle	0.769	0.561
Greedy (T=0)	**0.609**	0.374
Logprob	0.609	**0.397**
Ensemble	0.604	—
SJ	0.604	0.368
Vote	0.596	0.338
Random	0.580	0.338

log-probability is the only viable unsupervised signal—and only on tasks where greedy decoding is sufficiently suboptimal. For NER with fine-tuned models, greedy decoding is near-optimal and multi-sample selection adds no value with current signals.

Ensemble analysis. Combining signals via logistic regression yields NER AUROC 0.735 (+0.7pp over voting alone) and RE AUROC 0.719 (−0.2pp, no improvement). A linear combination of SJ and voting achieves optimal NER ρ=0.489 at α=0.3. FK receives a negative coefficient in the logistic model, indicating that surface agreement is net-negative when structural signals are present. These results suggest that voting captures nearly all discriminative information; ensemble gains are marginal.

5.4 Task-Complexity Boundary

All signals degrade from NER to RE (Table 1). SJ drops from ρ=0.357 (NER) to 0.211 (RE) on the full set—a −0.146 decline. This degradation reflects compound error propagation in relational tasks: RE requires correctly identifying both head and tail entities *and* the relation type. Errors in any component disrupt consistency measurement.

Conditional filtering (retaining only instances where $F_1 > 0$) substantially restores signal strength. On RE, SJ recovers from 0.211 to 0.368 (+0.157), and voting from 0.422 to 0.463 (+0.041). The conditional lift dwarfs temperature tuning effects ($\Delta\rho \leq \pm 0.01$ across T=0.7/1.0). This indicates that the NER–RE gap is primarily a *capability boundary* effect: consistency signals work well within the model's knowledge range but cannot diagnose systematic knowledge gaps.

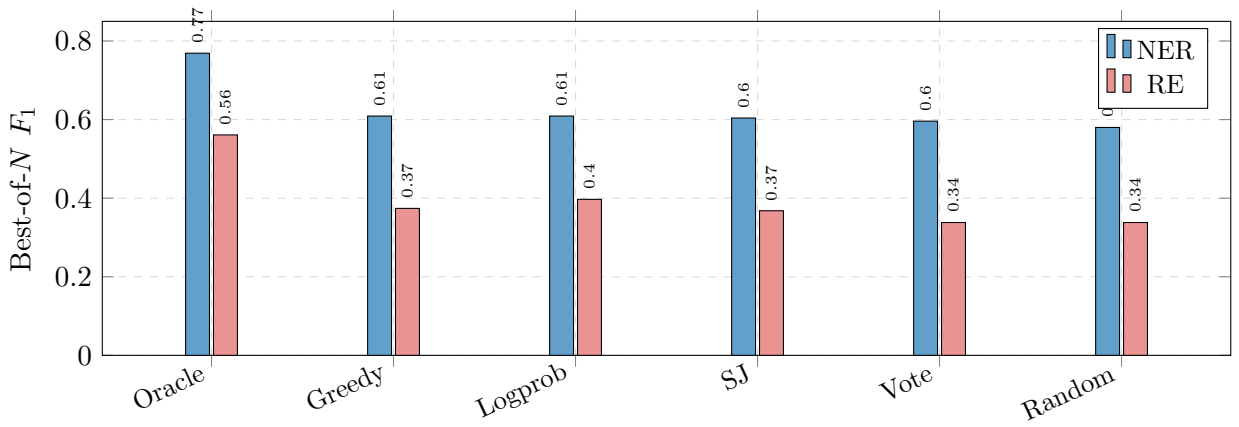


Figure 3: Best-of-N selection results on SciERC. On NER (blue), no signal surpasses greedy despite large oracle headroom (+16pp). On RE (red), only logprob exceeds greedy (+2.4pp). The gap between oracle and best achievable signal visualizes the *correlation–selection gap*.

Temperature analysis. SJ exhibits remarkable temperature robustness ($\Delta\rho = \pm 0.01$ between T=0.7 and T=1.0), while FK is sensitive ($\Delta\rho = \pm 0.06$–0.09). T=1.0 preserves more oracle headroom (RE: 0.167 vs. 0.131) and output diversity (1−SJ: 0.389 vs. 0.313). We recommend T=1.0 with conditional filtering over temperature optimization.

Sample budget scaling. Preliminary results (n=200 subset) suggest that N=8 → N=16 substantially improves RE signals (voting ρ: 0.340 → 0.448, +0.108) while NER saturates (voting ρ: 0.392 → 0.401). This is consistent with the task-complexity interpretation: compound error propagation increases per-instance variance, requiring more samples to average out. Full-set confirmation of this scaling pattern is left to future work.

6 Discussion

When does consistency outperform log-probability? Our results show a clear advantage for structural consistency under two conditions: (a) moderate model performance where log-probability calibration is unreliable, and (b) complex structured outputs where surface metrics fail. We expect structural signals to further outperform logprob when: (i) models are used in-context without fine-tuning (poorer calibration); (ii) log-probabilities are unavailable (closed-source APIs); (iii) the model is systematically overconfident. The OOD experiment (Section A) provides direct evidence: under domain shift, logprob correlation becomes *negative* ($\rho = -0.031$) while structural signals retain a positive, albeit weak, signal (ρ=0.078–0.093).

AUROC–ρ divergence. On LLaMA × CoNLL, logprob achieves the highest AUROC (0.761) but the lowest ρ (0.307) among structural signals. AUROC measures threshold classification ("is this above/below median quality?"), while ρ measures rank discrimination ("which is better than which?"). In the near-perfect regime (greedy F_1=0.927), logprob calibration is tight and thresholding works well. But for *ranking* instances—the prerequisite for both selective prediction and sample selection—structural signals remain superior. This underscores the importance of evaluating quality estimation with multiple metrics.

Limitations. Our study has several limitations. *Model scale*: We evaluate only 8B-parameter models; larger models (70B+) may exhibit different calibration properties that alter the structural–logprob balance. *Dataset coverage*: We evaluate NER on two datasets and RE on one (SciERC). Informal text domains (social media, conversational) are not covered; the OOD experiment (Section A) suggests substantial degradation in mismatched domains. *Selection gap*: We identify the correlation–selection gap but do not resolve it. Developing consistency-based per-sample selection signals (e.g., leave-one-out consistency scoring) is an important direction. *Supervised baselines*: We compare only unsupervised signals. Trained reward models or verifiers [Cobbe et al.(2021), Lightman et al.(2023)] likely outperform all unsupervised approaches; our contribution is the systematic comparison within the unsupervised regime.

7 Conclusion

We conducted a systematic comparison of five confidence signals for LLM-based information extraction across two models and two datasets. Structural consistency signals—entity voting confidence and Soft Jaccard—provide the strongest quality estimation, achieving AUROC 0.73 and ρ=0.50 on NER, substantially outperforming log-probability (AUROC 0.56, ρ=0.14). This advantage holds across a 2×2 model–dataset matrix with cross-model variation $|\Delta\rho| \leq 0.006$ and three-seed standard deviation $\sigma_\rho \leq 0.008$.

We identified a correlation–selection gap: despite strong instance-level quality prediction, no unsupervised consistency signal beats greedy decoding for NER best-of-N selection. Only per-sample log-probability enables selection on RE, where greedy decoding is suboptimal. This distinction between quality estimation ("which inputs are reliable?") and sample selection ("which output is best?") has direct implications for system design: use structural consistency for selective prediction/abstention, and log-probability for sample selection when applicable.

All signals degrade on relational tasks due to compound error propagation, but conditional filtering restores discriminative power. These findings provide actionable guidance for deploying multi-sample IE systems and highlight the need for per-sample structural consistency metrics to close the selection gap.

A Out-of-Domain Evaluation

To assess robustness under domain shift, we evaluate the SciERC-fine-tuned Qwen3-8B on the WNUT-2017 social media NER test set (n=1,287; n=689 valid after filtering; n=378 conditional). Entity type schemas are disjoint (SciERC: Task/Method/Material/Metric; WNUT: person/location/corporation/cr work), creating a challenging zero-shot transfer scenario.

Greedy F_1 drops from 0.66 (SciERC in-domain) to 0.143, with 61.4% of instances producing F_1=0. All quality estimation signals degrade substantially (Table 6). On the full set, all correlations are near zero ($\rho \leq 0.093$), with logprob turning negative (ρ=−0.031).

Two observations are noteworthy. First, the *relative ordering* of signals is preserved: voting > SJ > FK > EM > logprob, identical to the in-domain ranking. This suggests that structural consistency is not a dataset-specific artifact. Second, logprob's negative correlation reflects calibration failure under domain shift: the model assigns confidently high probabilities to incorrect entity types. Structural signals, which measure inter-sample agreement rather than model confidence, are immune to this miscalibration—they are weak but positive.

On the conditional set (excluding F_1=0 instances), signals recover partially (SJ ρ=0.198, voting ρ=0.224), following the same pattern as in-domain conditional analysis.

Table 6: OOD quality estimation: SciERC → WNUT-2017 (Qwen3-8B, N=8). All signals degrade substantially, but structural signals retain positive correlation while logprob turns negative.

	Full (n=689)		Conditional (n=378)	
Signal	ρ	AUROC	ρ	AUROC
VOTE	0.093	0.574	0.224	0.631
SJ	0.078	0.562	0.198	0.618
FK	0.064	0.548	0.152	0.594
EM	0.051	0.539	0.141	0.587
LOGPROB	−0.031	0.487	0.117	0.573

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